

COSC 594

Variational Inference (II)

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Announcement

Check Canvas for Homework 1

Due on 9/17, 11:59pm

Outline

- RECAP
- VARIATIONAL AUTOENCODERS
- VARIATIONAL DIFFUSION MODELS

Outline

■ RECAP

■ **RECAP** (continued)

■ **RECAP** (continued)

■ **RECAP** (continued)

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■ **RECAP** (continued)

■ **RECAP** (continued)

Latent variable model

Data set



$$(x, \textcolor{red}{z}) \sim q(\cdot, \cdot)$$

Generation



$$(x, \textcolor{red}{z}) \sim p(\cdot, \cdot)$$

$\textcolor{red}{z}$ – latent variable

■ JOINT DISTRIBUTION DIFFERENCE MINIMIZATION

$$\underset{p(\cdot, \cdot)}{\text{minimize}} \quad D_{\text{KL}}(q(\cdot, \cdot) \parallel p(\cdot, \cdot))$$

Evidence (likelihood) lower bound (ELBO)

$$\text{ELBO : } \text{ELBO}(p) := \mathbb{E}_{x \sim q(\cdot)} \mathbb{E}_{z \sim q(\cdot | x)} \left[\log \frac{p(x, z)}{q(z | x)} \right]$$

■ LOWER BOUND

$$\mathbb{E}_{x \sim q(\cdot)} [\log p(x)] \geq \text{ELBO}(p)$$

■ JOINT KL DIVERGENCE

$$D_{\text{KL}}(q(\cdot, \cdot) \| p(\cdot, \cdot)) = -\text{ELBO}(p) + \mathbb{E}_{x \sim q(\cdot)} [\log q(x)]$$

Sufficiency of ELBO maximization

Joint KL divergence minimization: $\underset{p(\cdot, \cdot)}{\text{minimize}} \quad D_{\text{KL}}(q(\cdot, \cdot) \parallel p(\cdot, \cdot))$

Joint likelihood maximization: $\underset{p(\cdot, \cdot)}{\text{maximize}} \quad \mathbb{E}_{(x, z) \sim q(\cdot, \cdot)} [\log p(x, z)]$

ELBO maximization: $\underset{p(\cdot, \cdot)}{\text{maximize}} \quad \text{ELBO}(p)$

■ CHAIN RULE FOR KL DIVERGENCE

$$\underset{p(\cdot, \cdot)}{\text{minimize}} \quad D_{\text{KL}}(q(\cdot, \cdot) \parallel p(\cdot, \cdot)) \quad \Rightarrow \quad \underset{p(\cdot)}{\text{minimize}} \quad D_{\text{KL}}(q(\cdot) \parallel p(\cdot))$$



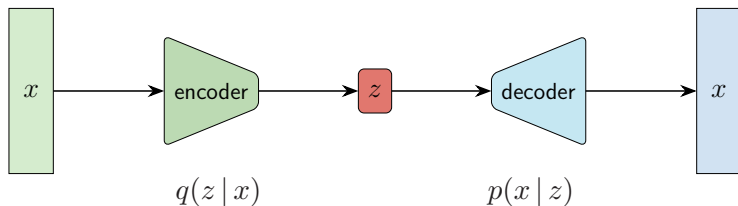
$$\underset{p(\cdot)}{\text{maximize}} \quad \mathbb{E}_{x \sim q(\cdot)} [\log p(x)]$$

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- RECAP

- VARIATIONAL AUTOENCODERS

Variational autoencoder



z – latent variable

■ LATENT VARIABLES

- ★ **characters** in images
- ★ **trajectory types** in robotic trajectories
- ★ **functional groupings** of protein sequences

Parametrization of ELBO

Encoder : $q_{\phi}(z | x)$ w/ parameter ϕ

Decoder : $p_{\theta}(x | z)$ w/ parameter θ

■ ELBO DECOMPOSITION

$$\begin{aligned}\text{ELBO} &= \mathbb{E}_{x \sim q(\cdot)} \mathbb{E}_{z \sim q_{\phi}(\cdot | x)} \left[\log \frac{p_{\theta}(x, z)}{q_{\phi}(z | x)} \right] \\&= \mathbb{E}_{x \sim q(\cdot)} \mathbb{E}_{z \sim q_{\phi}(\cdot | x)} \left[\log \frac{p_{\theta}(x | z) p(z)}{q_{\phi}(z | x)} \right] \\&= \mathbb{E}_{x \sim q(\cdot)} \mathbb{E}_{z \sim q_{\phi}(\cdot | x)} \left[\log p_{\theta}(x | z) + \log \frac{p(z)}{q_{\phi}(z | x)} \right] \\&= \underbrace{\mathbb{E}_{x \sim q(\cdot)} \mathbb{E}_{z \sim q_{\phi}(\cdot | x)} [\log p_{\theta}(x | z)]}_{\text{reconstruction}} - \underbrace{\mathbb{E}_{x \sim q(\cdot)} [D_{\text{KL}}(q_{\phi}(\cdot | x) \| p(\cdot))]}_{\text{prior matching}}\end{aligned}$$

ELBO maximization

$$\underset{\phi, \theta}{\text{maximize}} \quad \mathbb{E}_{x \sim q(\cdot)} \mathbb{E}_{z \sim q_{\phi}(\cdot | x)} \left[\log \frac{p_{\theta}(x | z) p(z)}{q_{\phi}(z | x)} \right]$$

$p(z) = \mathcal{N}(z; 0, I)$ – Gaussian prior

■ GAUSSIAN ENCODER

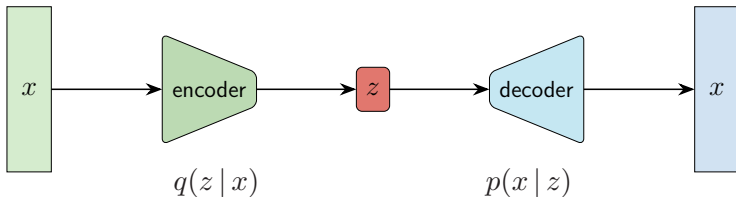
$$q_{\phi}(z | x) = \mathcal{N}\left(z; \mu_{\phi}(x), \sigma_{\phi}^2(x)I\right)$$

★ ‘observable’ latent

$$z = \mu_{\phi}(x) + \sigma_{\phi}(x) \circ \epsilon \quad \text{for } \epsilon \sim \mathcal{N}(0, I)$$

Tractable distribution gradient

Hierarchical variational autoencoder



$z = z_{1:T}$ – latent sequence

■ MARKOVIAN LATENT STRUCTURE

$$q(z_{1:T} | x) = q(z_1 | x)q(z_2 | z_1) \dots q(z_T | z_{T-1})$$

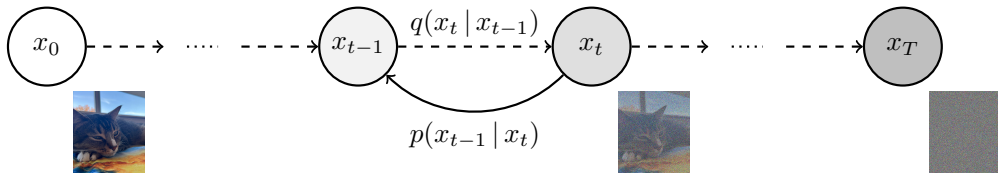
$$p(x | z_{1:T}) = p(z_T)p(z_{T-1} | z_T) \dots p(x | z_1)$$

ELBO maximization for recursive VAE

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Diffusion model



data point x_0 – image, trajectory, video, protein

■ ANALOGUE W/ VAE

- ★ input and latent

$$x = x_0 \quad \text{and} \quad z_{1:T} = x_{1:T}$$

- ★ noiser (encoder)

$$q(x_{1:T} | x_0) = q(x_1 | x_0)q(x_2 | x_1) \dots q(x_T | x_{T-1})$$

- ★ denoiser (decoder)

$$p(x_0 | x_{1:T}) = p(x_T)p(x_{T-1} | x_T) \dots p(x_0 | x_1)$$

ELBO for diffusion model

$$\mathbb{E}_{x_0 \sim q(\cdot)} \mathbb{E}_{x_{1:T} \sim q(\cdot | x_0)} \left[\log \frac{p(x_{1:T}, x_0)}{q(x_{1:T} | x_0)} \right] = \text{TERM}_1 + \text{TERM}_2 + \text{TERM}_3$$

★ reconstruction

$$\text{TERM}_1 = \mathbb{E}_{x_0} \mathbb{E}_{x_1 \sim q(\cdot | x_0)} [p(x_0 | x_1)]$$

★ prior matching

$$\text{TERM}_2 = -\mathbb{E}_{x_0} [D_{\text{KL}}(q(x_T | x_0) \| p(x_T))]$$

★ denoising matching

$$\text{TERM}_3 = -\sum_{t=2}^T \mathbb{E}_{x_0} \mathbb{E}_{x_t \sim q(\cdot | x_0)} [D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \| p(x_{t-1} | x_t))]$$

■ APPLICATION OF BAYES RULE

$$q(x_t | x_{t-1}, x_0) = \frac{q(x_{t-1} | x_t, x_0) q(x_t | x_0)}{q(x_{t-1} | x_0)}$$

$$\begin{aligned} \text{ELBO} &= \mathbb{E}_{x_0} \mathbb{E}_{x_{1:T} \sim q(\cdot | x_0)} \left[\log \frac{p(x_{1:T}, x_0)}{q(x_{1:T} | x_0)} \right] \\ &= \mathbb{E}_{x_0} \mathbb{E}_{x_{1:T} \sim q(\cdot | x_0)} \left[\log \frac{p(x_T) p(x_0 | x_1) \prod_{t=2}^T p(x_{t-1} | x_t)}{q(x_1 | x_0) \prod_{t=2}^T q(x_t | x_{t-1}, x_0)} \right] \\ &= \mathbb{E}_{x_0} \mathbb{E}_{x_{1:T} \sim q(\cdot | x_0)} \left[\log \frac{p(x_T) p(x_0 | x_1)}{q(x_1 | x_0)} + \sum_{t=2}^T \log \frac{p(x_{t-1} | x_t)}{q(x_t | x_{t-1}, x_0)} \right] \\ &= \mathbb{E}_{x_0} \mathbb{E}_{x_{1:T} \sim q(\cdot | x_0)} \left[\log \frac{p(x_T) p(x_0 | x_1)}{q(x_T | x_0)} + \sum_{t=2}^T \log \frac{p(x_{t-1} | x_t)}{q(x_{t-1} | x_t, x_0)} \right] \\ &= \mathbb{E}_{x_0} \mathbb{E}_{x_{1:T} \sim q(\cdot | x_0)} \left[\log p(x_0 | x_1) + \log \frac{p(x_T)}{q(x_T | x_0)} + \sum_{t=2}^T \log \frac{p(x_{t-1} | x_t)}{q(x_{t-1} | x_t, x_0)} \right] \end{aligned}$$

ELBO for diffusion model

$$\mathbb{E}_{x_0 \sim q(\cdot)} \mathbb{E}_{x_{1:T} \sim q(\cdot | x_0)} \left[\log \frac{p(x_{1:T}, x_0)}{q(x_{1:T} | x_0)} \right] = \text{TERM}_1 + \text{TERM}_2 + \text{TERM}_3$$

★ reconstruction

$$\text{TERM}_1 = \mathbb{E}_{x_0} \mathbb{E}_{x_1 \sim q(\cdot | x_0)} [p(x_0 | x_1)]$$

★ prior matching

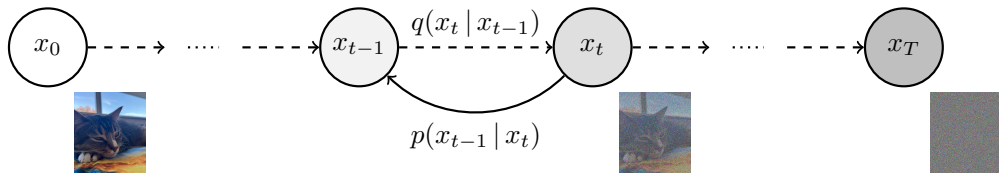
$$\text{TERM}_2 = -\mathbb{E}_{x_0} [D_{\text{KL}}(q(x_T | x_0) \| p(x_T))]$$

★ denoising matching

$$\text{TERM}_3 = -\sum_{t=2}^T \mathbb{E}_{x_0} \mathbb{E}_{x_t \sim q(\cdot | x_0)} [D_{\text{KL}}(q(\textcolor{red}{x}_{t-1} | \textcolor{red}{x}_t, \textcolor{red}{x}_0) \| \textcolor{blue}{p}(\textcolor{blue}{x}_{t-1} | \textcolor{blue}{x}_t))]$$

Difference between **ground-truth denoiser** and **approximate denoiser**?

Gaussian encoder



data point x_0 – image, trajectory, video, protein

■ REPARAMETERIZATION TRICK

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \underbrace{\sqrt{\alpha_t} x_{t-1}}_{\mu_t(x_t)}, \underbrace{(1 - \alpha_t) I}_{\Sigma_t(x_t)})$$

$\Updownarrow$

$$x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{1 - \alpha_t} \epsilon \quad \text{for } \epsilon \sim \mathcal{N}(0, I)$$

■ REPETITION OF REPARAMETERIZATION TRICK

$$\begin{aligned}x_t &= \sqrt{\alpha_t} \mathbf{x}_{t-1} + \sqrt{1 - \alpha_t} \epsilon \\&= \sqrt{\alpha_t} \sqrt{\alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{1 - \alpha_{t-1}} \epsilon + \sqrt{1 - \alpha_t} \epsilon \\&\vdots \\&= \sqrt{\prod_{i=1}^t \alpha_i} x_0 + \sqrt{1 - \prod_{i=1}^t \alpha_i} \epsilon \\&= \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon \\&\Leftrightarrow q(x_t | x_0) \sim \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) I)\end{aligned}$$

■ APPLICATION OF BAYES RULE

$$q(x_{t-1} | x_t, x_0) = \frac{q(x_t | x_{t-1}, x_0) q(x_{t-1} | x_0)}{q(x_t | x_0)}$$

■ GROUND-TRUTH DENOISER

$$q(x_{t-1} \mid x_t, x_0) = \mathcal{N}(x_{t-1}; \mu_q(x_t, x_0), \sigma_q^2(t)I)$$

★ mean

$$\mu_q(x_t, x_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t}x_0$$

★ variance

$$\sigma_q^2(t) = \frac{(1 - \alpha_t)(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}$$

Linear mean and constant variance

Denoising matching as mean estimation

$$\text{TERM}_3 = - \sum_{t=2}^T \mathbb{E}_{x_0} \mathbb{E}_{x_t \sim q(\cdot | x_0)} [D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \| p(x_{t-1} | x_t))]$$

★ ground-truth denoiser

$$q(x_{t-1} | x_t, x_0) = \mathcal{N}(x_{t-1}; \mu_q(x_t, x_0), \sigma_q^2(t)I)$$

★ approximate denoiser

$$p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma_q^2(t)I)$$

★ MEAN SQUARE MINIMIZATION

$$\underset{\theta}{\text{minimize}} \quad D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \| p_\theta(x_{t-1} | x_t)) \quad \Leftrightarrow \quad \underset{\theta}{\text{minimize}} \quad \|\mu_q(x_t, x_0) - \mu_\theta(x_t, t)\|^2$$

Denoising matching as ground truth prediction

- ★ mean of ground-truth denoiser

$$\mu_q(x_t, x_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t}x_0$$

- ★ mean of approximate denoiser

$$\mu_\theta(x_t, t) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1 - \alpha_t)}{1 - \bar{\alpha}_t}\hat{x}_\theta(x_t, t)$$

- ★ MEAN SQUARE MINIMIZATION

$$\underset{\theta}{\text{minimize}} \quad D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \parallel p_\theta(x_{t-1} | x_t)) \quad \Leftrightarrow \quad \underset{\theta}{\text{minimize}} \quad \|x_0 - \hat{x}_\theta(x_t, t)\|^2$$

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- VARIATIONAL AUTOENCODERS

 - Application of ELBO, recursive VAE

- VARIATIONAL DIFFUSION MODELS

 - ELBO maximization $\approx$ Denoising matching